Dual Nature of Radiation and Matter

Single Correct Type Questions

1. If a source of electromagnetic radiation having power 15kW produces 10^{16} photons per second, the radiation belongs to a part of spectrum is.

(Take Planck constant $h = 6 \times 10^{-34}$ Js)

[31 Jan, 2023 (Shift-I)]

(*a*) Micro waves (*b*) Ultraviolet rays
(*c*) Gamma rays (*d*) Radio waves

2. A 2 mW laser operates at a wavelength of 500 nm. The number of photons that will be emitted per second is [Given Planck's constant $h = 6.6 \times 10^{-34}$ Js,]

[10 April, 2019 (Shift-II)]

(*a*) 2×10^{16} (*b*) 1.5×10^{16}
(*c*) 5×10^{15} (*d*) 1×10^{16}

3. The work functions of Aluminium and Gold are 4.1 eV and 5.1 eV respectively. The ratio of the slope of the stopping potential versus frequency plot for Gold to that of Aluminium is **[6 April, 2023 (Shift-II)]**

(*a*) 1.24 (*b*) 2
(*c*) 1 (*d*) 1.5

4. The difference between threshold wavelengths for two metal surfaces A and B having work function $\phi_A = 9$ eV and $\phi_B = 4.5$ eV in nm is

(Given, hc = 1242 eV nm) **[13 April, 2023 (Shift-I)]**

(*a*) 264 (*b*) 138
(*c*) 276 (*d*) 540

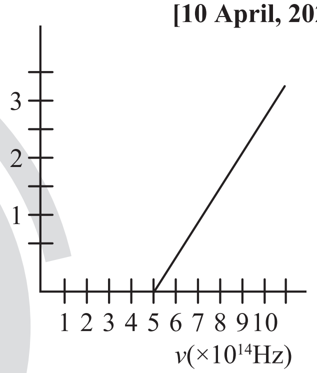
5. A metallic surface is illuminated with radiation of wavelength λ, the stopping potential is V_0. If the same surface is illuminated with radiation of wavelength 2λ, the stopping potential becomes $\frac{V_o}{4}$. The threshold wavelength for this metallic surface will be -

[11 April, 2023 (Shift-I)]

(*a*) $\frac{\lambda}{4}$ (*b*) 4λ (*c*) $\frac{3}{2}\lambda$ (*d*) 3λ

6. The variation of stopping potential (V_0) as a function of the frequency (ν) of the incident light for a metal is shown in figure. The work function of the surface is

[10 April, 2023 (Shift-II)]



(*a*) 18.6 *eV* (*b*) 2.98 *eV*
(*c*) 2.07 *eV* (*d*) 1.36 *eV*

7. The threshold wavelength for photoelectric emission from a material is 5500Å. Photoelectrons will be emitted, when this material is illuminated with monochromatic radiation from a

A. 75 *W* infra-red lamp B. 10 *W* infra-red lamp
C. 75 *W* ultra-violet lamp D. 10 *W* ultra-violet lamp

Choose the correct answer from the options given below:

[29 Jan, 2023 (Shift-I)]

(*a*) B and C only (*b*) A and D only
(*c*) C only (*d*) C and D only

8. The threshold frequency of metal is f_0. When the light of frequency $2f_0$ is incident on the metal plate, the maximum velocity of photoelectron is v_1. When the frequency of incident radiation is increased to $5f_0$. the maximum velocity of photoelectrons emitted is v_2. The ratio of v_1 to v_2 is: **[01 Feb, 2023 (Shift-II)]**

(*a*) $\frac{v_1}{v_2} = \frac{1}{2}$ (*b*) $\frac{v_1}{v_2} = \frac{1}{8}$

(*c*) $\frac{v_1}{v_2} = \frac{1}{16}$ (*d*) $\frac{v_1}{v_2} = \frac{1}{4}$

9. A metal exposed to light of wavelength 800 nm and and emits photoelectrons with a certain kinetic energy. The maximum kinetic energy of photo-electron doubles when light of wavelength 500 nm is used. The work function of the metal is: (Take hc = 1230 eV -nm).

[25 July, 2022 (Shift-I)]

(*a*) 1.537 eV (*b*) 2.46 eV
(*c*) 0.615 eV (*d*) 1.23 eV

10. Two streams of photons, possessing energies equal to five and ten times the work function of metal are incident on the metal surface successively. The ratio of maximum velocities of the photoelectron emitted, in the two cases respectively, will be **[28 July, 2022 (Shift-II)]**

(*a*) 1 : 2 (*b*) 1 : 3 (*c*) 2 : 3 (*d*) 3 : 2

11. The light of two different frequencies whose photons have energies 3.8 eV and 1.4 eV respectively, illuminate a metallic surface whose work function is 0.6 eV successively. The ratio of maximum speed of emitted electrons for the two frequencies respectively will be:

[24 June, 2022 (Shift-II)]

(*a*) 1 : 1 (*b*) 2 : 1 (*c*) 4 : 1 (*d*) 1 : 4

12. In a photoelectric experiment ultraviolet light of wavelength 280 nm is used with lithium cathode having work function ϕ = 2.5eV. If the wavelength of incident light is switched to 400nm, find out the change in the stopping potential. ($h = 6.63 \times 10^{-34}$ Js, $c = 3 \times 10^{8} ms^{-1}$)

[26 Aug, 2021 (Shift-I)]

(*a*) 1.3 V (*b*) 1.1 V
(*c*) 0.6 V (*d*) 1.9 V

13. When the wavelength of radiation falling on a metal is charged from 500 nm to 200 nm the maximum kinetic energy of the photoelectrons becomes three times larger. The work function of the metals is close to:

[03 Sep, 2020 (Shift-I)]

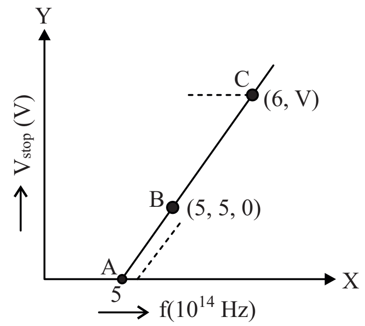
(*a*) 1.02 eV (*b*) 0.61 eV
(*c*) 0.52 eV (*d*) 0.81 eV

14. Radiation, with wavelength 6561 Å falls on a metal surface to produce photoelectrons. The electrons are made to enter a uniform magnetic field of 3×10^{-4} T. If the radius of the largest circular path followed by the electrons is 10 mm, the work function of the metal is close to

[09 Jan, 2020 (Shift-I)]

(*a*) 1.8 eV (*b*) 0.8 eV
(*c*) 1.6 eV (*d*) 1.1 eV

15. Given figure shows few data points in a photo electric effect experiment for a certain metal. The minimum energy for ejection of electron from its surface is (Planck's constant h = 6.62×10^{-34} Js) **[04 Sep, 2020 (Shift-I)]**

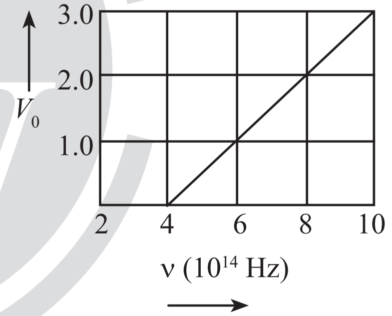


(*a*) 2.59 eV (*b*) 2.27 eV
(*c*) 1.93 eV (*d*) 2.10 eV

16. The stopping potential V_0 (in volt) as a function of frequency (v) for a sodium emitter, is shown in the figure. The work function of sodium, from the data plotted in the figure, will be : **[12 April, 2019 (Shift-I)]**

(Given : Planck's constant (h) = 6.63×10^{-34} Js, electron charge $e = 1.6 \times 10^{-19} C$)

(*a*) 1.82e V (*b*) 1.66e V
(*c*) 2.12e V (*d*) 195e V



17. Surface of certain metal is first illuminated with light of wavelength λ_1 = 350 nm and then the light of wavelength λ_2 = 540 nm. It is found that the maximum speed of the photo electrons in the two cases differ by a factor of 2. The work function of the metal (in eV) is close to

[9 Jan, 2019 (Shift-I)]

(Energy of photon = $\dfrac{1240}{\lambda(\text{in nm})} eV$)

(*a*) 1.8 (*b*) 2.5 (*c*) 5.6 (*d*) 1.4

18. The magnetic field associated with a light wave is given, at the origin, by

$B = B_0 [\sin(3.14 \times 10^7)ct + \sin(6.28 \times 10^7)ct]$

If this light falls on a silver plate having a work function of 4.7 eV, what will be the maximum kinetic energy of the photo electrons? **[9 Jan, 2019 (Shift-II)]**

(*a*) 6.82 eV (*b*) 12.5 eV
(*c*) 8.52 eV (*d*) 7.67 eV

19. Electron beam used in an electron microscope, accelerated by a voltage of 20kV has a de-Broglie wavelength of λ_0. If the voltage is increased to 40kV then the de-Broglie wavelength associated with the electron beam would be: **[25 Jan, 2023 (Shift-I)]**

(a) $3\lambda_0$ (b) $9\lambda_0$

(c) $\frac{\lambda_0}{2}$ (d) $\frac{\lambda_0}{\sqrt{2}}$

20. The ratio of wavelengths of proton and deuteron accelerated by potential V_p and V_d is 1: $\sqrt{2}$. Then, the ratio of V_p to V_d will be: **[25 July, 2022 (Shift-II)]**

(a) 1 : 1 (b) $\sqrt{2}$:1

(c) 2 : 1 (d) 4 : 1

21. An α particle and a proton are accelerated from rest through the same potential difference. The ratio of linear momenta acquired by above two particals will be: **[29 July, 2022 (Shift-II)]**

(a) $\sqrt{2}:1$ (b) $2\sqrt{2}:1$

(c) $4\sqrt{2}:1$ (d) 8 : 1

22. An α particle and a carbon 12 atom has same kinetic energy K. The ratio of their de-Broglie wavelengths (λ_α : λ_{C12}) is: **[27 June, 2022 (Shift-I)]**

(a) $1:\sqrt{3}$ (b) $\sqrt{3}:1$

(c) 3 : 1 (d) $2:\sqrt{3}$

23. An electron of mass m and a photon have same energy E. The ratio of wavelength of electron to that of photon is : (c being the velocity of light)**[17 March, 2021 (Shift-I)]**

(a) $\frac{1}{c}\left(\frac{E}{2m}\right)^{1/2}$

(b) $\left(\frac{E}{2m}\right)^{1/2}$

(c) $c(2mE)^{\frac{1}{2}}$

(d) $\frac{1}{c}\left(\frac{2m}{E}\right)^{1/2}$

24. An electron moving with speed v and a photon moving with speed c, have same de-Broglie wavelength. The ratio of kinetic energy of electron to that of photon is: **[25 July, 2021 (Shift-II)]**

(a) $\frac{v}{2c}$ (b) $\frac{2c}{v}$ (c) $\frac{v}{3c}$ (d) $\frac{3c}{v}$

25. The de-Broglie wavelength of a particle having kinetic energy E is λ. How much extra energy must be given to this particle so that the de-Broglie wavelength reduces to 75% of the initial value? **[26 Aug, 2021 (Shift-II)]**

(a) $\frac{1}{9}E$ (b) E

(c) $\frac{16}{9}E$ (d) $\frac{7}{9}E$

26. Particle A of mass $m_A = \frac{m}{2}$ moving along the x-axis with velocity v_0 collides elastically with another particle B at rest having mass $m_B = \frac{m}{3}$. If both particles move along the x-axis after the collision, the change $\Delta\lambda$ in de-Broglie wavelength of particle A, in terms of its de-Broglie wavelength (λ_0) before collision is **[04 Sep, 2020 (Shift-I)]**

(a) $\Delta\lambda = 2\lambda_0$ (b) $\Delta\lambda = 4\lambda_0$

(c) $\Delta\lambda = \frac{3}{2}\lambda_0$ (d) $\Delta\lambda = \frac{5}{2}\lambda_0$

27. A particle moving with kinetic energy E has de-Broglie wavelength λ. If energy ΔE is added to its energy, the wavelength become $\lambda/2$. Value of ΔE, is **[9 Jan, 2020 (Shift-I)]**

(a) $4E$ (b) $3E$

(c) $2E$ (d) E

28. An electron (mass m) with initial velocity $\vec{v} = v_0\hat{i} + v_0\hat{j}$ is in an electric field $\vec{E} = -E_0\hat{k}$. If λ_0 is initial de-Broglie wavelength of electron, its de-Broglie wavelength at time t is given by **[08 Jan, 2020 (Shift-II)]**

(a) $\frac{\lambda_0}{\sqrt{1+\frac{e^2E_0^2t^2}{2m^2v_0^2}}}$

(b) $\frac{\lambda_0}{\sqrt{2+\frac{e^2E^2t^2}{m^2v_0^2}}}$

(c) $\frac{\lambda_0\sqrt{2}}{\sqrt{1+\frac{e^2E^2t^2}{m^2v_0^2}}}$

(d) $\frac{\lambda_0}{\sqrt{1+\frac{e^2E^2t^2}{m^2v_0^2}}}$

29. Two particles move at right angle to each other. Their de-Broglie wavelengths are λ_1 and λ_2 respectively. The particles suffer perfectly inelastic collision. The de-Broglie wavelength λ, of the final particle, is given by

[08 April, 2019 (Shift-I)]

(*a*) $\lambda = \dfrac{\lambda_1 + \lambda_2}{2}$ (*b*) $\dfrac{2}{\lambda} = \dfrac{1}{\lambda_1} + \dfrac{1}{\lambda_2}$

(*c*) $\lambda = \sqrt{\lambda_1 \lambda_2}$ (*d*) $\dfrac{1}{\lambda^2} = \dfrac{1}{\lambda_1^2} + \dfrac{1}{\lambda_2^2}$

30. A particle A of mass '*m*' and charge '*q*' is accelerated by a potential difference of 50 V. Another particle B of mass '4 *m*' and charge '*q*' is accelerated by a potential difference of 2500 V. The ratio of de-Broglie wavelength $\dfrac{\lambda_A}{\lambda_B}$ is close to

[12 Jan, 2019 (Shift-I)]

(*a*) 10.00 (*b*) 0.07

ANSWER KEY

1. (*c*)	**2.** (*c*)	**3.** (*c*)	**4.** (*b*)	**5.** (*d*)	**6.** (*c*)	**7.** (*d*)	**8.** (*a*)	**9.** (*c*)	**10.** (*c*)
11. (*b*)	**12.** (*a*)	**13.** (*b*)	**14.** (*d*)	**15.** (*b*)	**16.** (*b*)	**17.** (*a*)	**18.** (*d*)	**19.** (*d*)	**20.** (*d*)
21. (*b*)	**22.** (*b*)	**23.** (*a*)	**24.** (*a*)	**25.** (*d*)	**26.** (*b*)	**27.** (*b*)	**28.** (*a*)	**29.** (*d*)	**30.** (*c*)

EXPLANATIONS

1. (*c*) Energy of one photon $E = \dfrac{\text{Power}}{\text{Photon frequency}}$

$= \dfrac{15\times10^{3}}{10^{16}}$

$\because E = hv$

$\Rightarrow v = \dfrac{15\times10^{-13}}{6\times10^{-34}} = 2.5\times10^{21}$

So the radiation belongs to gamma Rays.

2. (*c*) $P = \dfrac{n.hc}{\lambda}$

$n = \dfrac{p\lambda}{h.c}$

$= \dfrac{2\times10^{-3}\times500\times10^{-9}}{6.6\times10^{-34}\times3\times10^{8}} = 5\times10^{15}$

3. (*c*) $eV_s = K_{max} = hf - \phi$

$V_s = \left(\dfrac{h}{e}\right)f + \left(\dfrac{-\phi}{e}\right)$

Slope of stopping potential is independent of nature of metal.

Hence, $\text{slope}(V_s)_{\text{Gold}} = \text{slope}(V_s)_{\text{Aluminium}}$

4. (*b*) Formula of work function $\phi = \dfrac{hc}{\lambda}$

$\lambda_A = \left(\dfrac{1242}{9}\right) = 138 \text{ nm}$

$\lambda_B = \left(\dfrac{1242}{4.5}\right) = 276 \text{ nm}$

$\lambda_B - \lambda_A = 138 \text{ nm}$

5. (*d*) From the equation of photoelectric effect

$eV_0 = \dfrac{hc}{\lambda} - \dfrac{hc}{\lambda_0}$

$\dfrac{eV_0}{4} = \dfrac{hc}{2\lambda} - \dfrac{hc}{\lambda_0}$

Pluggin in the value of eV_O

$\Rightarrow \dfrac{1}{4}\left(\dfrac{hc}{\lambda} - \dfrac{hc}{\lambda_0}\right) = \dfrac{hc}{2\lambda} - \dfrac{hc}{\lambda_0}$

$\dfrac{1}{4}\dfrac{hc}{\lambda} - \dfrac{2hc}{2\lambda} = \dfrac{hc}{4\lambda_0} - \dfrac{hc}{\lambda_0}$

$\dfrac{3}{4\lambda_0} = \dfrac{1}{4\lambda}$

$\Rightarrow \lambda_0 = 3\lambda$

6. (*c*) $eV_0 = hv - \phi$

$\therefore$ work function, $\phi = hv$

$= 6.6 \times 10^{-34} \times 5 \times 10^{14} = 33 \times 10^{-20}$ J

Stopping potential will be zero when incident and threshold frequencies are equal.

$\phi_{(eV)} = \dfrac{33\times10^{-20}}{1.6\times10^{-19}} = 2.07eV$

7. (*d*) As the threshold wavelength is 5500Å, therefore for photoelectric emission wavelength of radiation,

$\lambda < 5500$Å

and wavelength of UV radiation,

$\lambda_{uv} < 5500$Å

$\therefore$ option (*d*) is correct

8. (*a*) $\because K_{max} = hf - hf_0$

For incident radiation having fequency = $2f_0$

$\dfrac{1}{2}mv_1^2 = 2hf_0 - hf_0 = hf_0$

For incident radiation having fequency = $5f_0$

$\dfrac{1}{2}mv_2^2 = 5hf_0 - hf_0 = 4hf_0$

$\Rightarrow \left(\dfrac{v_1}{v_2}\right) = \dfrac{1}{4} \Rightarrow \dfrac{v_1}{v_2} = \dfrac{1}{2}$

9. (*c*) Einstein's equation of photo-electric current

$$K.E = \frac{hc}{\lambda} - \phi$$

$$\therefore\ K \cdot E_1 = \frac{1230}{800} - \phi \quad ...(i)$$

$$K \cdot E_2 = 2K \cdot E_1 = \frac{1230}{500} - \phi \quad ...(ii)$$

From (*i*) & (*ii*)

$$2 \times \frac{1230}{800} - 2\phi = \frac{1230}{500} - \phi$$

$$\phi = \frac{1230}{400} - \frac{1230}{500}$$

$$\phi = 0.615 \text{ eV}$$

10. (*c*) $E_1 = 5\phi - \phi = 4\phi = \frac{1}{2}mv_1^2$

$$E_2 = 10\phi - \phi = \frac{1}{2}mv_2^2$$

$$\Rightarrow \frac{V_1}{V_2} = \frac{\sqrt{\frac{8\phi}{m}}}{\sqrt{\frac{18\phi}{m}}}\sqrt{\frac{8}{18}} = \frac{\sqrt{4}}{9} = \frac{2}{3}$$

11. (*b*) $KE_1 = h\upsilon_1 - \phi_0$

$$\frac{1}{2}mv_1^2 = 3.8 - 0.6 = 3.2\text{eV} \quad ...(i)$$

Similarly,

$$KE_2 = h\upsilon_2 - \phi_0$$

$$\frac{1}{2}mv_2^2 = 1.4 - 0.6 = 0.8\text{eV} \quad ...(ii)$$

On dividing equation (i) with equation (ii)

$$\left(\frac{v_1}{v_2}\right)^2 = 4 \Rightarrow \frac{v_1}{v_2} = 2:1$$

12. (*a*) Given, $\lambda_1 = 280$ nm

$\lambda_2 = 400$ nm

$\phi = 2.5\ eV.$

$$\frac{hc}{\lambda_1} = \phi + eV_1$$

$$hc\left(\frac{1}{\lambda_1} - \frac{1}{\lambda_2}\right) = e(V_1 - V_2)$$

$$V_1 - V_2 = \frac{hc}{e}\left(\frac{1}{\lambda_1} - \frac{1}{\lambda_2}\right)$$

$$= 12400\left(\frac{1}{2800} - \frac{1}{4000}\right)\left[\lambda_1 \text{ and } \lambda_2 \text{ are taken in } \mathring{A}\right]$$

$= 4.4 - 3.1$

$= 1.3$V

13. (*b*) K.E. of emitted photo electron,

$$KE = \frac{hc}{\lambda} - \phi \quad ...(i)$$

$\therefore$ At $\lambda_1 = 500$ nm

$$KE_1 = \frac{hc}{\lambda_1} - \phi \quad ...(ii)$$

At $\lambda_2 = 200$ nm

$$KE_2 = \frac{hc}{\lambda_2} - \phi = 3K_1 \text{ (Given)}$$

$$= 3\left(\frac{hc}{\lambda_1} - \phi\right) \Rightarrow \frac{hc}{1}\left(\frac{3}{\lambda_1} - \frac{1}{\lambda_2}\right) = 2\phi$$

$$\Rightarrow \phi = \frac{hc}{2}\left(\frac{3}{\lambda_1} - \frac{3}{\lambda_2}\right)$$

$$= \frac{hc}{2 \times 100\,nm}\left(\frac{3}{5} - \frac{1}{2}\right) = \frac{1240}{2 \times 100}\frac{1}{10} = 0.62\,eV.$$

14. (*d*) $KE_{max} = E - \phi \Rightarrow K.E_{max} = \frac{12400}{\lambda(\text{in Å})} - \phi$ (in eV)

$$\therefore\ r = \frac{\sqrt{2mKE}}{eB}$$

$$KE_{max} = \frac{r^2e^2B^2}{2m} \text{ (in J)}$$

$$= \frac{r^2eB^2}{2m} \text{ (in eV)}$$

$$\therefore\ \phi = \frac{12400}{6561} - \frac{r^2eB^2}{2m} = 1.1\ eV$$

15. (*b*) $V = \frac{h}{e}f - \frac{\phi}{e}$

Minimum energy for ejection = work function

$\phi = hf$ (for $v = 0$)

$$= \frac{6.62 \times 10^{-34} \times 5.5 \times 10^{14}}{1.6 \times 10^{-19}}$$

$= 2.27$ eV

16. (*b*) $eV_0 = hv - \phi$

$$V_0 = \frac{hv - \phi}{e}$$

When $v = 4 \times 10^{14}\ Hz \Rightarrow V_0 = 0\ V$

$$\therefore \phi = hv = \frac{6.63 \times 10^{-34} \times 4 \times 10^{14}}{1.6 \times 10^{-19}} eV$$

$\phi = 1.66\ eV$

17. (*a*) $\frac{1240}{350} - \phi = (KE)_I = 4x$ ...(*i*)

$\frac{1240}{540} - \phi = (KE)_{II} = x$ ...(*ii*)

Equation (*i*) – Equation (*ii*)

$$\frac{1240}{350} - \frac{1240}{540} = 3.542 - 2.296 = 3x$$

$\Rightarrow 1.246 = 3x \Rightarrow x = 0.41$

$\therefore \phi = 2.296 - 0.41 = 1.886$

18. (*d*) The maximum kinetic energy of the photoelectrons,

$KE_{\max} = hv_{\max} - \phi$

[Here ϕ, is work function]

$$\frac{(6.6 \times 10^{-34})(6.28 \times 10^7)(3 \times 10^8)}{1.6 \times 10^{-19} \times 2 \times 3.14} - 4.7$$

$= 12.37 - 4.7 = 7.67\ eV$

19. (*d*) $\therefore \lambda = \frac{h}{\sqrt{2meV}}$

$$\therefore \lambda \,\alpha\, \frac{1}{\sqrt{V}} \quad \Rightarrow \frac{\lambda}{\lambda_0} = \sqrt{\frac{20}{40}}$$

$$\Rightarrow \lambda = \frac{\lambda_0}{\sqrt{2}}$$

20. (*d*) From de-Broglie hypothesis

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2meV}}$$

$$\Rightarrow \frac{\lambda_P}{\lambda_D} = \frac{\sqrt{2m_D eV_D}}{\sqrt{2m_P eV_P}} = \frac{1}{\sqrt{2}}$$

$$\Rightarrow \frac{2m_P V_D}{m_P V_P} = \frac{1}{2}$$

$$\Rightarrow \frac{V_P}{V_D} = 4:1 \qquad (\because m_D = 2\ m_p)$$

21. (*b*) $k_\alpha = 2eV \Rightarrow P_\alpha = \sqrt{2K_\alpha m_\alpha}$

$k_P = eV \Rightarrow P_P = \sqrt{2K_P m_P}$

$$\therefore \frac{P_\alpha}{P_P} = \sqrt{\frac{2 \times (4m_P)}{m_P}} = \sqrt{8}:1 = 2\sqrt{2}:1$$

22. (*b*) $\lambda = \frac{h}{P} = \frac{h}{\sqrt{2m(K.E)}}$

$\because$ K.E. = Constant

$$\therefore \lambda \propto \frac{1}{\sqrt{m}}$$

$$\Rightarrow \frac{\lambda_\alpha}{\lambda_{C12}} = \sqrt{\frac{m_{C12}}{m_\alpha}} = \sqrt{\frac{12}{4}} = \sqrt{3}:1$$

23. (*a*) For electron, $\lambda_1 = \frac{h}{mv} = \frac{h}{\sqrt{2mE}}$ and For photon, $\lambda_2 = \frac{hc}{E}$

$$\Rightarrow \frac{\lambda_1^2}{\lambda_2^2} = \frac{\frac{h^2}{2Em}}{\frac{h^2c^2}{E^2}} = \frac{E}{2mc^2} \Rightarrow \frac{\lambda_1}{\lambda_2} = \frac{1}{c}\left(\frac{E}{2m}\right)^{\frac{1}{2}}$$

24. (*a*) For electron,

$$(KE)_e = \frac{1}{2}mv^2 = \frac{pv}{2}$$

For photon,

$$(K.E)_p = \frac{hc}{\lambda} = pc \qquad \left(p = \frac{h}{\lambda}\right)$$

$$\frac{KE_e}{KE_p} = \frac{pv/2}{pc} = \frac{v}{2c}$$

25. (*d*) According to the question,

$$\lambda' = \lambda \times \frac{75}{100} = \frac{3}{4}\lambda$$

We know that,

$$\lambda = \frac{h}{P} = \frac{h}{\sqrt{2mE}}$$

$$\Rightarrow \frac{E'}{E} = \left(\frac{\lambda}{\lambda'}\right)^2 = \left(\frac{4}{3}\right)^2$$

$$\Rightarrow E' = \frac{16}{9}E$$

Therefore, extra energy required

$$= E' - E = \frac{16}{9}E - E = \frac{7}{9}E$$

26. (b) $\lambda_0 = \dfrac{h}{\dfrac{m}{2}v_0}\dfrac{m}{2}v_0 = \dfrac{m}{2}v_1 + \dfrac{m}{3}v_2$

$v_0 = v_2 - v_1$

$v_1 = \dfrac{v_0}{5}$

$\lambda_1 = \dfrac{h}{\dfrac{m}{2}\dfrac{v_0}{5}} = 5\lambda_0$

$\Delta\lambda = 4\lambda_0$

27. (b) $\lambda = \dfrac{h}{\sqrt{2mE}} \Rightarrow \lambda \propto \dfrac{1}{\sqrt{E}}$

$\dfrac{\lambda}{\lambda/2} = \sqrt{\dfrac{E_f}{E_i}}$

$4E_i = E_f$

$\Rightarrow \Delta E = 4E_i - E_i = 3E$

28. (a) Initially $m\left(\sqrt{2}v_0\right) = \dfrac{h}{\lambda_0}$

Velocity, $v = \sqrt{v_0^2 + v_0^2 + \left(\dfrac{eE_0}{m}t\right)^2} = \sqrt{2v_0^2 + \dfrac{e^2E_0^2}{m^2}t^2}$

$= v_0\hat{i} + v_0\hat{j} + \dfrac{eE_0}{m}t\hat{k}$

So wavelength, $\lambda = \dfrac{h}{mv} = \dfrac{h}{m\sqrt{2v_0^2 + \dfrac{e^2E_0^2}{m^2}t^2}}$

$\lambda = \dfrac{\lambda_0}{\sqrt{1 + \dfrac{e^2E_0^2}{2m^2v_0^2}t^2}}$

29. (d) $\dfrac{h}{\lambda_1} = P_1$ $\quad P_2 = \dfrac{h}{\lambda_2}$

$\vec{P}_1 = \dfrac{h}{\lambda_1}\hat{i}$ & $\vec{P}_2 = \dfrac{h}{\lambda_2}\hat{j}$

Using momentum conservation

$\vec{P} = \vec{P}_1 + \vec{P}_2 = \dfrac{h}{\lambda_1}\hat{i} + \dfrac{h}{\lambda_2}\hat{j}$

$|\vec{P}| = \sqrt{\left(\dfrac{h}{\lambda_1}\right)^2 + \left(\dfrac{h}{\lambda_2}\right)^2}$

$\dfrac{h}{\lambda} = \sqrt{\left(\dfrac{h}{\lambda_1}\right)^2 + \left(\dfrac{h}{\lambda_2}\right)^2}$

$\dfrac{1}{\lambda^2} = \dfrac{1}{\lambda_1^2} + \dfrac{1}{\lambda_2^2}$

30. (c) $\lambda = \dfrac{h}{\sqrt{2mqV}}$ $\quad \dfrac{\lambda_P}{\lambda_\alpha} = \sqrt{\dfrac{4\times1\times2500}{1\times1\times50}} = 10\sqrt{2} = 14.14$